A Band-of-Inaction Model of the Indian No-Arbitrage Band

Theoretical Framework for the BS Project — *Candidate Model B*

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Abstract

This is the second of two independently-developed theoretical frameworks for the project (the first being the constrained-arbitrage model, Candidate A). Where Candidate A builds the band up from an arbitrageur's optimisation, Candidate B takes the *reduced-form* route: it treats correcting a mispricing as a costly control problem and derives the no-arbitrage band as the *optimal region of inaction*, in the tradition of transaction-cost impulse-control models (Dumas, 1992; Constantinides). The pay-off of this route is that the band it produces is *exactly* the object our econometrics estimates: the basis is a regulated process that behaves as a random walk inside the band and mean-reverts only at its edges, which is a three-regime threshold model (the TAR / Ters–Urban TVECM). The document derives the band and its (asymmetric) edges from smooth-pasting conditions, gives their comparative statics, shows how the MWPL position limit and its "F&O ban" remove the correcting barrier, proves the discrete-time bridge to the threshold estimator, and states the cross-sectional and identification implications — so that, as with Candidate A, the project's hypotheses are theorems derived before any data work.

1 Purpose and how this differs from Candidate A

Both frameworks target the same object — the no-arbitrage band around a stock's cost-of-carry fair value — but from opposite directions. Candidate A models *who* corrects mispricing (a risk-averse arbitrageur) and derives the band from their constrained trade-off; its strength is the behavioural mechanism and the explicit shadow price of the ban. Candidate B abstracts from the identity of the arbitrageur and asks only: *given that correction is costly, when is it optimal to act?* Its strength is a razor-tight link to the estimator — the model's band *is* the threshold band we fit — and clean comparative statics from a single control problem. Read together they triangulate the same predictions; this document is written to stand alone as a candidate final framework.

2 Environment

Consider one F&O stock. As in Candidate A, let $F^* = Se^{(r-\delta)\tau}$ be the cost-of-carry fair futures value and let the (log) basis be

$$z_t \;=\; f_t - s_t - (r-\delta)\tau, \qquad f = \ln F,\; s = \ln S, \tag{1}$$

the stationary error-correction term of the cointegrated pair (f, s) (a fact confirmed for Indian single-stock futures by Roy & Chakraborty, 2023). *In the absence of arbitrage* — i.e. while no one is trading against the gap — order flow from hedgers, retail and noise traders pushes the basis around as a driftless diffusion,

$$\mathrm{d}z_t = \sigma\,\mathrm{d}W_t, \tag{2}$$

with W a standard Brownian motion and $\sigma > 0$ the instantaneous basis volatility. Equation (2) encodes the null of "no correction": left alone, the basis neither reverts nor is pinned to zero.

Costly correction. Pulling the basis back toward zero requires putting on the cash-and-carry trade (for $z > 0$: sell futures, buy spot) or the reverse cash-and-carry (for $z < 0$: buy futures, short spot). Each intervention carries a cost. We adopt the canonical impulse-control cost structure: an intervention that resets the basis incurs a fixed cost K_j that differs by side,

$$K_+ = K \quad \text{(upper side, } z > 0\text{)}, \qquad K_- = K + K_s \quad \text{(lower side, } z < 0\text{)}, \tag{3}$$

where $K > 0$ is the symmetric round-trip cost of establishing and unwinding the arbitrage program and $K_s > 0$ is the *extra* cost of the leg that must short the stock (SLB borrow fee, search, recall risk) — the same asymmetry as in Candidate A. While the basis is away from zero the arbitrageur bears a running "carry/opportunity" cost θz^2 per unit time ($\theta > 0$): capital tied to an open gap, financing and mark-to-market risk that scale with the squared deviation.[1] Future costs are discounted at rate $\rho > 0$.

The MWPL limit. As in Candidate A, the correcting position is capped: $|q| \le L = \kappa \min(0.15\,\mathrm{FF}, 65\,\mathrm{ADDV})$, and the "ban" (open interest $\ge 95\%$ of MWPL) forbids fresh positions. In the control language this raises — and in the limit makes infinite — the cost of the intervention that would reflect the basis, as formalised in §5.

3 The optimal band of inaction

The arbitrageur chooses an intervention policy to minimise the expected discounted sum of running and intervention costs. This is an impulse-control problem whose solution is a *barrier (band) policy*: do nothing while z lies in an interval $[C_1, C_2]$, and intervene to reset the basis toward zero the instant z reaches an edge. Formally the value function $V(z)$ satisfies the

[1] The quadratic running cost is the standard tractable choice (linear-quadratic control); the qualitative results below hold for any convex running cost.

quasi-variational inequality

$$\min\Big\{\ \rho V(z) - \tfrac{\sigma^2}{2}V''(z) - \theta z^2,\ \ V(z) - \big[\,V(0) + K_j\,\big]\ \Big\} = 0, \tag{4}$$

where the first term is the Hamilton–Jacobi–Bellman operator on the inaction region and the second is the intervention (obstacle) term. Inside the inaction region the HJB holds with equality:

$$\rho V(z) = \tfrac{\sigma^2}{2}V''(z) + \theta z^2. \tag{5}$$

Proposition 1 (Existence and shape of the no-arbitrage band)**.** *The solution of* (4) *is a band policy with inaction region* $[C_1, C_2]$*. On the inaction region the value function is*

$$V(z) = A_+ e^{\eta z} + A_- e^{-\eta z} + \frac{\theta}{\rho}z^2 + \frac{\theta\sigma^2}{\rho^2}, \qquad \eta \equiv \frac{\sqrt{2\rho}}{\sigma}, \tag{6}$$

and each edge C_j *is pinned down by a* value-matching *and a* smooth-pasting *condition,*

$$V(C_j) = V(0) + K_j, \qquad V'(C_j) = 0. \tag{7}$$

The band is asymmetric: because $K_- = K + K_s > K_+ = K$*, the lower edge is farther from zero,* $|C_1| > |C_2|$.

Proof. Equation (5) is a linear second-order ODE; its homogeneous solutions are $e^{\pm\eta z}$ with $\eta = \sqrt{2\rho}/\sigma$, and a particular solution for the forcing θz^2 is $(\theta/\rho)z^2 + \theta\sigma^2/\rho^2$ (substitute to verify: $\rho[(\theta/\rho)z^2 + \theta\sigma^2/\rho^2] = \theta z^2 + \theta\sigma^2/\rho$ and $\frac{\sigma^2}{2}\cdot 2\theta/\rho = \theta\sigma^2/\rho$, matching). Hence (6). Optimality of a barrier policy for a diffusion with convex running cost and fixed intervention cost is standard: value-matching says the arbitrageur is indifferent between acting and not at the edge (acting costs K_j and resets the state to 0), and smooth-pasting $V'(C_j) = 0$ is the first-order optimality (high-contact) condition selecting the edge. Since $K_- > K_+$, a larger deviation is tolerated before the more expensive lower-side intervention is triggered, so $|C_1| > |C_2|$. □

Closed form for a symmetric benchmark. To exhibit the comparative statics transparently, take the symmetric case ($K_j = K$, so $C_2 = -C_1 = b$) and solve (7) for small intervention cost. With $A_+ = A_- = A$ (symmetry), $V(z) = 2A\cosh(\eta z) + (\theta/\rho)z^2 + \theta\sigma^2/\rho^2$. Smooth pasting $V'(b) = 0$ gives $2A\eta\sinh(\eta b) = -2\theta b/\rho$, and value-matching $V(b) - V(0) = K$ gives $2A(\cosh(\eta b) - 1) + (\theta/\rho)b^2 = K$. Eliminating A and expanding for small ηb (using $(\cosh x - 1)/\sinh x = \tanh(x/2) \approx x/2 - x^3/24$) yields, at leading order,

$$\boxed{\ b\ =\ \left(\frac{6K\sigma^2}{\theta}\right)^{1/4}.\ } \tag{8}$$

Derivation of (8). From smooth-pasting, $A = -\theta b/[\rho\eta\sinh(\eta b)]$. Substitute into value-matching: $-2\theta b/[\rho\eta\sinh(\eta b)]\cdot(\cosh(\eta b) - 1) + \theta b^2/\rho = K$. Using $(\cosh\varepsilon - 1)/\sinh\varepsilon = \tanh(\varepsilon/2) \approx \varepsilon/2 - \varepsilon^3/24$ with $\varepsilon = \eta b$: the first term $\approx -2\theta b/(\rho\eta)\,[\eta b/2 - (\eta b)^3/24] = -\theta b^2/\rho + \theta\eta^2 b^4/(12\rho)$. Adding $\theta b^2/\rho$ leaves $\theta\eta^2 b^4/(12\rho) = K$. With $\eta^2 = 2\rho/\sigma^2$ this is $\theta b^4/(6\sigma^2) = K$, i.e. $b = (6K\sigma^2/\theta)^{1/4}$. □

The asymmetric edges follow the same formula with the side-specific cost: $C_2 = (6K\sigma^2/\theta)^{1/4}$ and $|C_1| = (6(K+K_s)\sigma^2/\theta)^{1/4} > C_2$.

Mapping to the empirics. Proposition 1 and (8) give requirements (1) and (2): an explicit, asymmetric no-arbitrage band $[C_1, C_2]$, wider on the short-constrained (negative-basis) side. This is Chapter 2's/Chapter 4's object, and $[C_1, C_2]$ is exactly what the threshold estimator reports (§6).

4 Comparative statics

Proposition 2 (The band widens with cost and volatility)**.** *For the band half-width* (8)*,*

$$\frac{\partial b}{\partial K} > 0, \qquad \frac{\partial b}{\partial \sigma^2} > 0, \qquad \frac{\partial b}{\partial \theta} < 0, \tag{9}$$

and the asymmetry $|C_1| - |C_2| = (6\sigma^2/\theta)^{1/4}\left[(K+K_s)^{1/4} - K^{1/4}\right]$ *is positive and increasing in the short-side cost* K_s *and in volatility* σ^2*. Thus stocks that are more volatile, costlier to arbitrage, or harder to short have wider and more asymmetric bands.*

Proof. Immediate from (8): $b \propto K^{1/4}\sigma^{1/2}\theta^{-1/4}$ gives the three signs; the asymmetry expression substitutes the side-specific costs, and $\partial/\partial K_s > 0$, $\partial/\partial\sigma^2 > 0$ follow since $(K+K_s)^{1/4} - K^{1/4} > 0$ is increasing in K_s and the prefactor increases in σ^2. □

Mapping to the empirics. This is requirement (2) sharpened and the cross-sectional engine (§7). The constraint index (Chapter 1) is, in this model, a proxy for the composite $(K, K_s, \sigma^2, \theta)$ that sets the band width; the reforms enter through K, K_s (§5).

5 The MWPL limit and the ban

The position cap enters through the intervention cost. Establishing the correcting position of size needed to reflect the basis is cheap when the cap is slack, but as market-wide open interest approaches the MWPL the marginal position becomes unavailable, so the *effective* intervention cost rises; at the ban it is prohibitive. Model this by letting the side-specific cost depend on cap utilisation $\omega \equiv \text{OI/MWPL} \in [0, 1]$,

$$K_j(\omega) = K_j \cdot g(\omega), \qquad g'(\omega) > 0, \qquad g(\omega) \to \infty \text{ as } \omega \to 0.95^-, \tag{10}$$

capturing that the correcting trade gets dearer as the limit nears and becomes impossible at the ban.

Proposition 3 (The ban removes the reflecting barrier)**.** *Under* (10)*, the band half-width on side* j *is* $b_j(\omega) = (6K_j g(\omega)\sigma^2/\theta)^{1/4}$*, which is increasing in utilisation* ω *(*$\partial b_j/\partial\omega > 0$*). As* $\omega \to 0.95^-$*,* $b_j \to \infty$*: the reflecting edge recedes to infinity, the basis ceases to be regulated, and* (2) *governs it unchecked — z becomes a non-stationary driftless martingale on that side. Hence entering the ban discretely widens the band and eliminates mean reversion until the ban lifts.*

Proof. Substitute $K_j g(\omega)$ for K_j in (8): $b_j(\omega) = (6K_j g(\omega)\sigma^2/\theta)^{1/4}$, increasing in g hence in ω; as $g \to \infty$, $b_j \to \infty$. With no finite reflecting edge the QVI obstacle in (4) never binds, so z solves $\mathrm{d}z = \sigma\,\mathrm{d}W$ without intervention, a martingale with no stationary distribution. □

Proposition 3 is Candidate B's version of the ban result: instead of a shadow price jumping (Candidate A), the *reflecting barrier vanishes*. The two descriptions are the same phenomenon — correction switches off — seen through the two modelling lenses. Because the constrained leg is the short side, the negative edge C_1 blows up first and fastest, deepening the pre-existing negative-basis asymmetry.

Mapping to the empirics. Requirement (3): the ban widens the band discretely and kills reflection. The 2024–26 reforms act through $g(\cdot)$ and through the level of K_j (e.g. the ADDV-based MWPL changes what ω means), giving the dated shifts the TVECM event-study reads off (Chapter 4).

6 The discrete-time bridge to the threshold estimator

The reason to build this model is that its solved object maps one-to-one onto the econometrics.

Proposition 4 (The regulated basis is a three-regime threshold model)**.** *Sample the controlled basis at interval Δ. Inside the band, z evolves by (2) with no intervention, so*

$$z_t = z_{t-1} + \varepsilon_t, \qquad \varepsilon_t \sim (0, \sigma^2\Delta), \qquad z_{t-1} \in (C_1, C_2), \tag{11}$$

a unit-root (no-correction) process. Once z reaches an edge it is pushed back toward the interior, which in discrete time appears as mean reversion with a negative error-correction loading in the outer regimes:

$$\Delta z_t = \begin{cases} \alpha_-\,(z_{t-1} - C_1) + \varepsilon_t, & z_{t-1} \le C_1, \\ \varepsilon_t, & C_1 < z_{t-1} < C_2, \\ \alpha_+\,(z_{t-1} - C_2) + \varepsilon_t, & z_{t-1} \ge C_2, \end{cases} \qquad \alpha_-, \alpha_+ < 0. \tag{12}$$

Equation (12) is exactly the three-regime threshold autoregression / threshold VECM estimated in the project (Yadav–Pope–Paudyal; Dwyer et al.; Ters–Urban), with thresholds (C_1, C_2) equal to the model's band edges and a middle regime that is (near) a random walk.

Proof. Inside the inaction region there is no intervention, so z inherits (2), giving (11) with a unit root. At the barriers the reflection of a regulated diffusion contributes a local mean-reverting drift; discretely sampled, this is captured by a negative loading on the (demeaned) deviation in the outer regimes, yielding (12). The regime boundaries are the barriers C_1, C_2 by construction. □

Mapping to the empirics. Proposition 4 is the payoff of Candidate B and covers requirement (4): the estimator is not an arbitrary functional form but the *exact* discrete image of the model, so the fitted Ters–Urban thresholds *are* estimates of the structural band edges, and the near-zero middle-regime error-correction coefficient is the model's inaction region. It also tells us how to read a ban: a stretch during which the outer-regime loading collapses toward zero and the effective band widens.

7 Cross-sectional pricing

Proposition 5 (Wider bands, larger and more persistent mispricing, higher signal returns)**.** *On the inaction region the basis is a martingale, so absent intervention its dispersion grows with time; the stationary dispersion of z under the band policy is increasing in the band width b. Therefore a basis-signal strategy (trade toward correction when z exits the band) has expected return increasing in b, and hence — by Proposition 2 — increasing in K, K_s, σ^2 and in cap tightness. Cross-sectionally, high-constraint stocks exhibit larger average $|z|$, longer half-lives, and higher signal returns.*

Proof. A wider inaction interval lets z wander further before reflection, raising the variance of its stationary law (for regulated Brownian motion on $[C_1, C_2]$ the stationary distribution spreads with the interval length). Larger typical $|z|$ and rarer/weaker reflection raise both the frequency and the size of profitable signal trades, so expected signal return rises in b; the dependence on primitives follows from Proposition 2. □

Mapping to the empirics. Requirement (5): Chapter 5's Fama–MacBeth prediction that signals pay more where constraints bind, obtained here from the width of the inaction region.

8 Identification

Proposition 6 (Exogenous cost/limit variation shifts the band)**.** *The band edges $C_j = (6K_j g(\omega)\sigma^2/\theta)^{1/4}$ depend on the position limit through $\omega = OI/\text{MWPL}$ and through K_j. Because $\text{MWPL} = \kappa \min(0.15\,\text{FF}, 65\,\text{ADDV})$ depends on free float and* trailing, quarterly-reset *ADDV, exogenous movements in these denominators shift ω (and hence C_j) independently of the contemporaneous order-flow innovations ε_t driving z. This provides an instrument for ban status / band width, with first stage $\partial C_j/\partial\text{MWPL} < 0$ (a larger limit lowers ω, narrowing the band).*

Proof. C_j is increasing in ω (Proposition 3) and ω is decreasing in MWPL for given OI, so $\partial C_j/\partial\text{MWPL} < 0$. Free float and trailing ADDV are predetermined relative to ε_t, delivering the exclusion restriction $\text{Cov}(\text{MWPL}_t, \varepsilon_t) = 0$ by timing. □

Mapping to the empirics. Requirement (6): the same ADDV/free-float instrument as Candidate A and the project document, here shifting the band width rather than a shadow price. The 95% RD is again a supporting check subject to McCrary bunching.

9 Mapping to the empirical design

10 Assumptions, limitations, and relation to Candidate A

The model's tractability rests on three choices: a driftless-diffusion basis (2) (the "no-correction" null), a quadratic running cost, and fixed-cost impulse control. Relaxing them changes constants, not signs: a mean-reverting fundamental component adds drift inside the band (still a threshold model, now with a small inner loading); a *proportional* intervention cost c replaces impulse control with a reflecting (singular-control) band whose width scales as $b \propto (c\sigma^2/\theta)^{1/3}$ (Dumas,

Requirement	**Model result**	**Empirical counterpart**
(1) Explicit band	Prop. 1; $b = (6K\sigma^2/\theta)^{1/4}$	Ch. 2 basis; Ch. 4 band
(2) Asymmetry	Prop. 1–2: $\lvert C_1\rvert > \lvert C_2\rvert$ via K_s	Aggarwal $-15/+3$ bp; H1
(3) Ban widens band	Prop. 3: barrier $\to \infty$	Ch. 3 causal; H1/H4
(4) Slower correction / estimator	Prop. 4: TAR/TVECM is the discrete image	TVECM regimes; PEAD; H2
(5) Cross-section	Prop. 5: return $\uparrow$ in b	Fama–MacBeth; H5
(6) Identification	Prop. 6: $\partial C_j/\partial \text{MWPL} < 0$, exogenous	IV via ADDV/FF; H3/H6

Table 1: Model B delivers every requirement, with an exact bridge to the threshold estimator.

1992) — same comparative statics; and heterogeneous costs across arbitrageurs smooth the edges into a transition band, i.e. the STAR generalisation. What the reduced-form approach does *not* provide is the behavioural mechanism or the welfare/relative-returns content — for those, Candidate A (and the DSSW/Shleifer–Vishny apparatus behind it) is needed. That is the sense in which the two frameworks are complements: **Candidate A supplies the mechanism and the shadow price of the ban; Candidate B supplies the reduced-form band that the econometrics estimates directly.** Both reach, from primitives and before any data work, the same six predictions — which is exactly the pre-registration discipline this stage is meant to enforce.

Bottom line. Treating arbitrage as costly control, the no-arbitrage band emerges as an optimal region of inaction whose width and asymmetry follow $b = (6K\sigma^2/\theta)^{1/4}$-type formulas, whose constrained edge blows up under the ban, and whose discrete-time image *is* the threshold VECM we estimate. The project's hypotheses are again theorems, obtained independently of Candidate A.